**Question 1**

Determine the time complexity of the following code and express your answer in Big-O notation.

```
for(i = 1; i <= n; i++)
{
   for(j = 1; j <= n; j = j * 2)
   {
      for(k = 1; k <= i; k++)
      {
         print("*");
      }
   }
}
```

**Question 2**

Prove or disprove the following statements:

a) $5n^2 + 3n + 10 \in \Theta(n^2)$
b) $2^n \in O(n^3)$

**Question 3**

Solve the following recurrence using **Master Method**:

$$T(n) = 2T(n/2) + n\log n$$

**Question 4**

Solve the following recurrence using **Recursive Tree Method**:

$$T(n) = 3T(n/2) + n$$